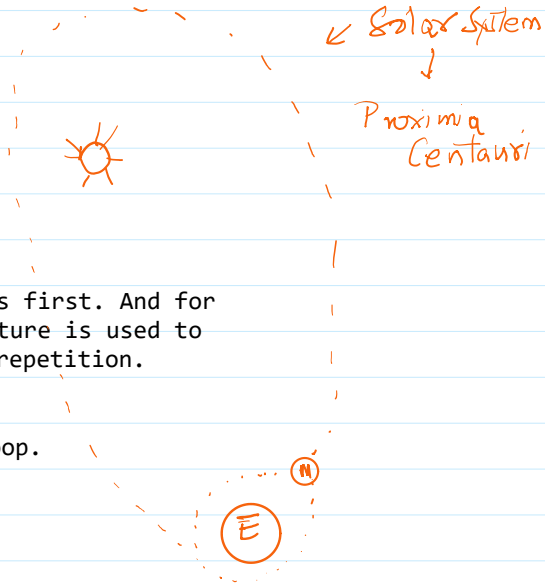
Nested for loop

04 October 2025 09:48

Syntax:

```
for (initialization; condition; increment/update){

    //outer loop statements
    for (initialization; condition; increment/update){
       //Inner or nested loop statements
    }
    //outer loop statements
}
```



A nested for loop is a loop inside another for loop. The outer loop runs first. And for each iteration of the outer loop, the inner loop runs fully. This structure is used to handle multi-dimensional data or task that requires multiple layers of repetition.

Structure of Nested for loops

- Outer loop controls the number of complete executions of the inner loop.
- Inner loop runs completely for each iteration of the outer loop.

start = 6
end = 10
times = 10
p = 1 2 3 ... 10 11

```
public class NestedLoop_Q1
{
    public static void main(String args[]){
        int start = 5, end = 10, p, times = 10;
        System.out.print("\nMultiplication table from 5 to 10 for 10 times.");

        for(;start<=end; start++){
            for(p = 1; p <= times; p++){
                System.out.print("\n" + p +" * " + start + " = " + (p * start));
            }
            System.out.println("\n================");
        }
    }
}
```

1 * 5 = 5
2 * 5 = 10
3 * 5 = 15

10 * 5 = 50

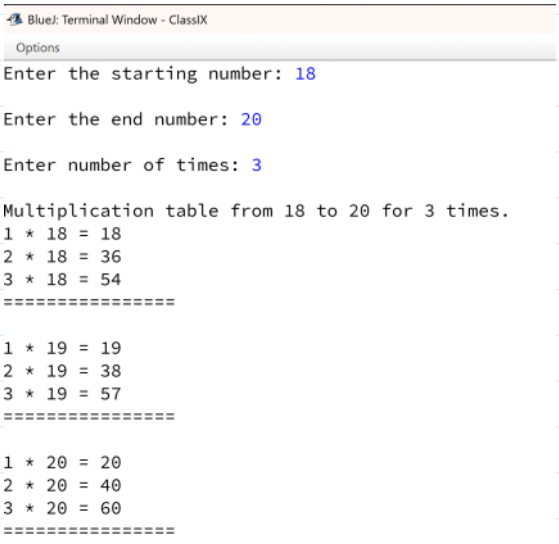
```
package NestedLoops;
import java.util.*;

/**
 * Write a description of class NestedLoop_Q1 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class NestedLoop_Q1
{
    public static void main(String args[]){
        System.out.print("\f");
        int start = 5, end = 10, p, times = 15;

        Scanner sc = new Scanner(System.in);
        System.out.print("\fEnter the starting number: ");
        start = sc.nextInt();
        System.out.print("\nEnter the end number: ");
        end = sc.nextInt();
        System.out.print("\nEnter number of times: ");
        times = sc.nextInt();
```

```
        System.out.print("\nMultiplication table from "+start +
            " to " + end + " for " + times + " times.");
        for(;start<=end; start++){
            for(p = 1; p <= times; p++){
                System.out.print("\n" + p +" * " + start + " = " + (p * start));
            }
            System.out.println("\n================");
        }
    }
}
```

Output:

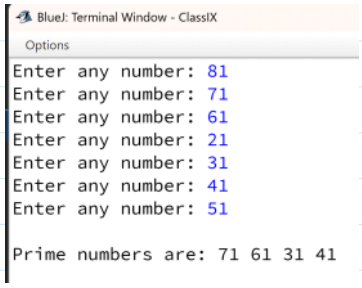



```
package NestedLoops;
import java.util.*;


/**
 * Write a description of class NestedLoop_Q2 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class NestedLoop_Q2
{
    public static void main(String[] args){
        System.out.print("\f");
        Scanner sc = new Scanner(System.in);
        String output="";
        int num, start = 1, end = 7, d, count;
        for(;start<=end; start++){
            System.out.print("Enter any number: ");
            num = sc.nextInt();
            //loop to detect num is prime or not
            for(count = 0, d = 1; d <= num; d++){
                if( num % d == 0 ){
                    count++;
                }
            }
            if(count == 2){
                //concatenation or joining of values one after other.
                output += num + " ";
            }
        }
```

```
        System.out.print("\nPrime numbers are: " + output);
    }
}
```

Output:



$\frac{10}{24}\overline{)100}\overline{)\cdot 416}$

96

40

24

160

144

16

$n = 3$

$S =$

$$\frac{1+2}{1\times 2} + \frac{1+2+3}{1\times 2\times 3} + \frac{1+2+3+4}{1\times 2\times 3\times 4}$$

$= 1.500 + 1 + 0.416$

≈ 2.916

```
package NestedLoops;
import java.util.*;


/**
 * Write a description of class NestedLoop_Q3 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class NestedLoop_Q3
{
    public static void main(String[] args){
        System.out.print("\f");
        Scanner sc = new Scanner(System.in);
        int n, nsum, pdeno, p, q;
        double frac, sum = 0;

        System.out.print("Enter number of terms: ");
        n = sc.nextInt();

        for(p = 1; p <= n; p++){
            nsum = 0; pdeno = 1;
            for(q = 1; q <= (p+1); q++){
                nsum += q;
                pdeno *= q;
            }
            frac = (double)nsum/pdeno;
            sum += frac;
        }
        System.out.print("\nS = " + sum);
    }
}
```

Output:

```
BlueJ: Terminal Window - ClassIX
Options
Enter number of terms: 3

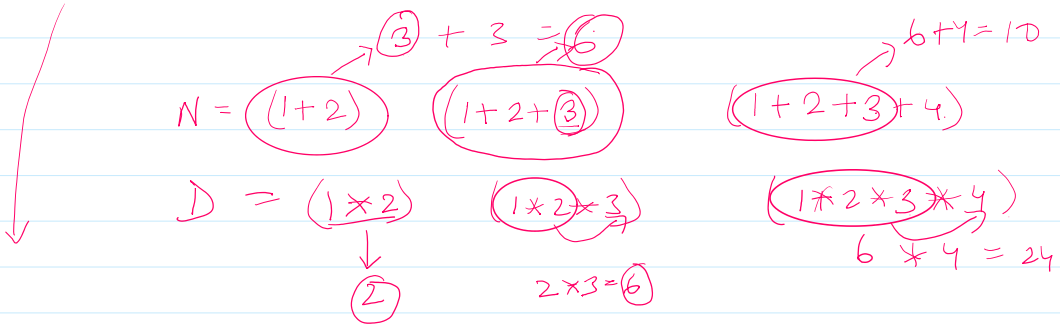
S = 2.9166666666666665
```

Home work:

3. Write a program to compute and display the sum of the following series: [ICSE-2007]

Without nested loop is also possible.

$$S = \frac{1+2}{1*2} + \frac{1+2+3}{1*2*3} + \text{----------} + \frac{1+2+3\text{----------}+n}{1*2*3\text{----------}*n}$$



print the series along with sum.

n = 3

→ output: (1+2)/(1*2) + (1+2+3)/(1*2*3) + (1+2+3+4)/(1*2*3*4) = 2.91666.....

4. Write a program to calculate and display the factorials of all the numbers between 'm' and 'n' (where m<n, m>0, n>0).
[Hint: factorial of 5 means: 5!=5*4*3*2*1]

m = 3, n = 10 ← data input

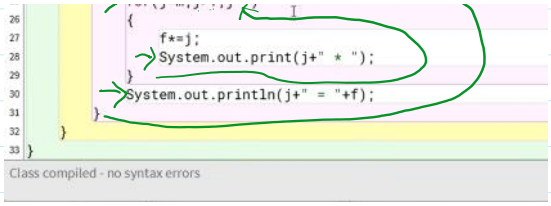
```
for( ; m<n; m++){
 int f=1; System.out.print(m+"!=")
    for( j=m; j>1 ; j--){
        f=f*j;
        System.out.print(j+"*");
    }
     System.out.print(j + "=" +f);
}
```

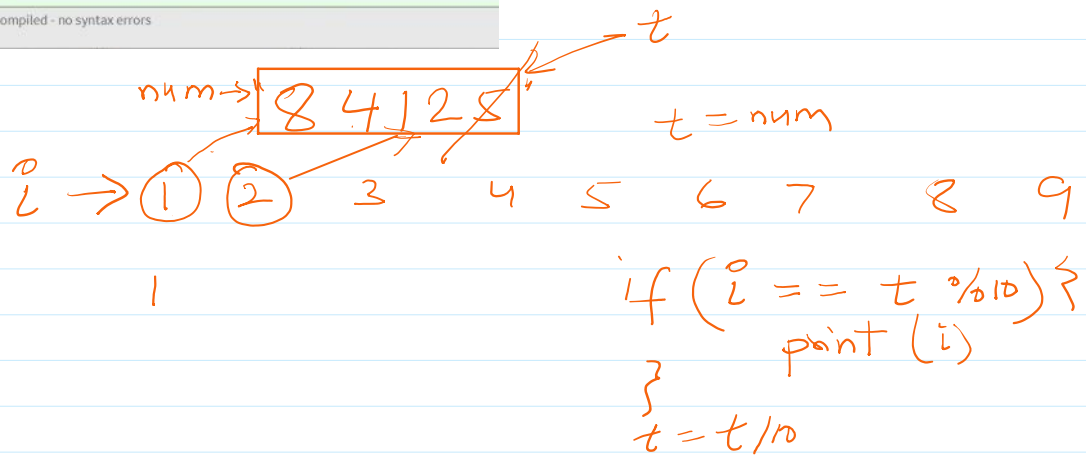
```
10 import java.util.Scanner;
11 public class Pg281_Q4
12
13 {
14     public static void main(String[] args)
15     {
16         Scanner sc = new Scanner(System.in);
17         System.out.print("\fEnter lesser no.: ");
18         int m = sc. nextInt();
19         System.out.print("\nEnter greater no.: ");
20         int n = sc. nextInt();
21         int f=1, j;
22         for(;m<=n;m++)
23         {
24             System.out.print("\n" + m + "! = ");
25             for(j=m;j>1;j--)
26             {
27                 f*=j;
28                 System.out.print(j+" * ");
29             }
30             System.out.println(j+" = "+f);
```

n = 7
m = 3
f = 1 2 6
j = 3 2 1

"\n" + m + "! = " Output:

3! = 3 * 2 * 1 = 6






```
package NestedLoops;
import java.util.*;


/**
 * Write a description of class Q13b here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q13b
{
    public static void main(String args[]){
        //1 + (1 + 2) + (1 + 2 + 3) + ...+(1+2+3+...+n)
        int n, i, s = 0, t = 0;
        System.out.print("\f");
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of terms: ");
        n = sc.nextInt();
        for(i = 1; i <= n; i++){
          t +=i;
          s += t;
        }
      System.out.print("\nS = " + s);
    }
}
```

1 + 3 + 6 + 10
1 + (1+2) + (1+2+3) + (1+2+3+4) = 20

n = 4
i = ~~X 2 3 4~~ 5
t = ~~0 X 3 6~~ 10
s = ~~0 X 1 4 10~~ 20

S = 20

```
package NestedLoops;
import java.util.*;

/**
 * Write a description of class Q13b_A here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q13b_A
{
    public static void main(String args[]){
        //print the series with sum
        //1 + (1 + 2) + (1 + 2 + 3) + ...+(1+2+3+...+n)
        int n, i, s = 0, t = 0, j;
```

1 2 3 4 5
2 3 4 5
3 4 5

```
        //print the series with sum
        //1 + (1 + 2) + (1 + 2 + 3) + ...+(1+2+3+...+n)
        int n, i, s = 0, t = 0, j;
        System.out.print("\f");
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of terms: ");
        n = sc.nextInt();
        for(i = 1; i <= n; i++){
            t +=i;
            s += t;
            System.out.print("(");
            for(j = 1; j < i; j++){
                System.out.print(j + " + ");
            }

            if(i < n){
                System.out.print(j + ") + ");
            }
            else{
                System.out.print(j + ") = "+s);
            }

        }
    }
}
```

2 3 4 5
3 4 5
4 5
5

```
package NestedLoops;

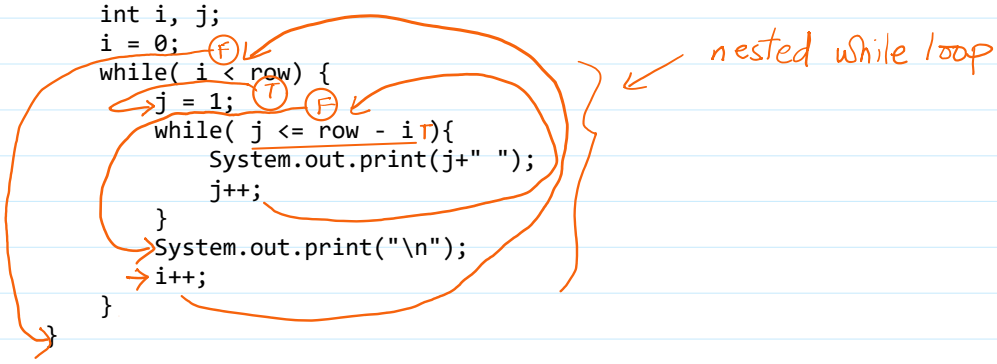

/**
 * Write a description of class Q14_bPg282 here.
 * String.format("%3d", j)
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q14_bPg282
{
    public static void main(String args[]){
        System.out.print("\f");
        int row = 25;
        int i, j;
        i = 0;
        while( i < row) {
            for(j = 1; j <= row - i; j++){
                System.out.print(j+" ");
            }
            System.out.print("\n");
            i++;
        }
    }
}
```

nested for loop inside while loop.

```
package NestedLoops;

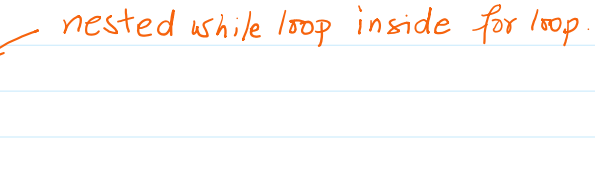

/**
 * Write a description of class Q14_bPg282 here.
 * String.format("%3d", j)
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q14_bPg282
```

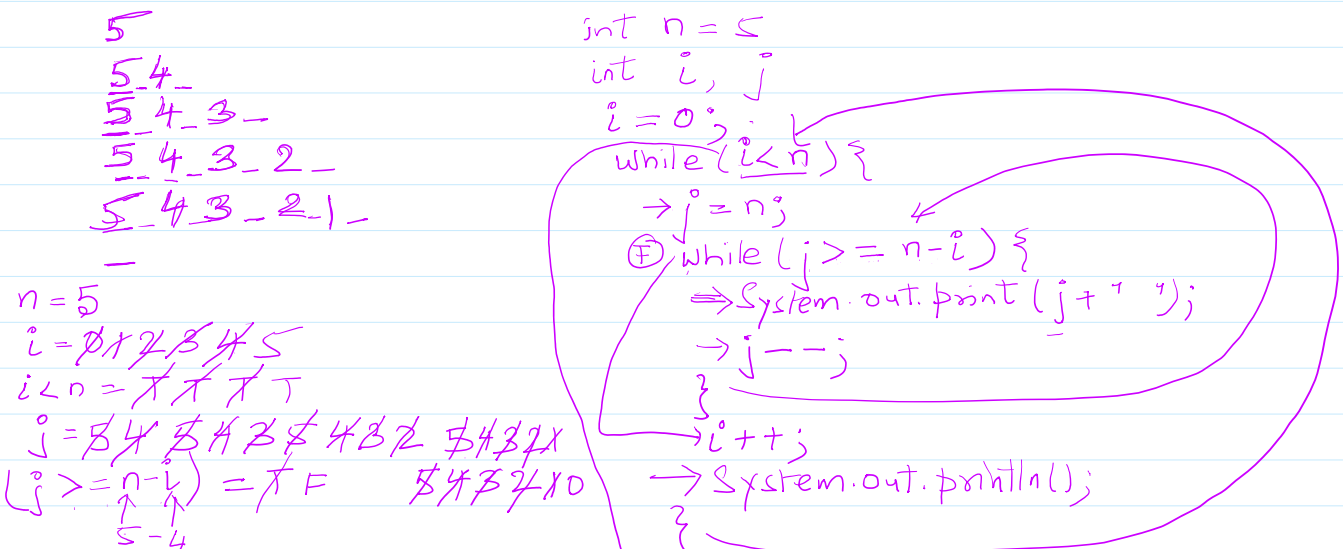
```
{
    public static void main(String args[]){
        System.out.print("\f");
        int row = 25;
        int i, j;
        i = 0;
        while( i < row) {
            j = 1;
            while( j <= row - i ){
                System.out.print(j+" ");
                j++;
            }
            System.out.print("\n");
            i++;
        }
    }
}
```

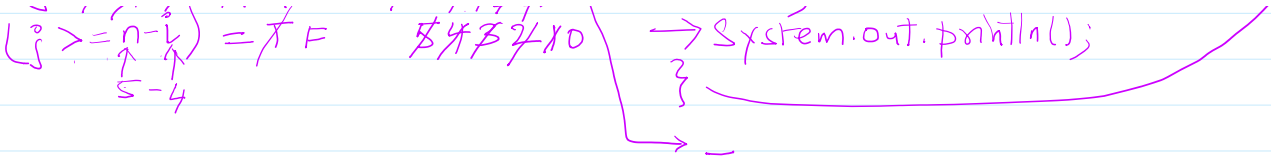


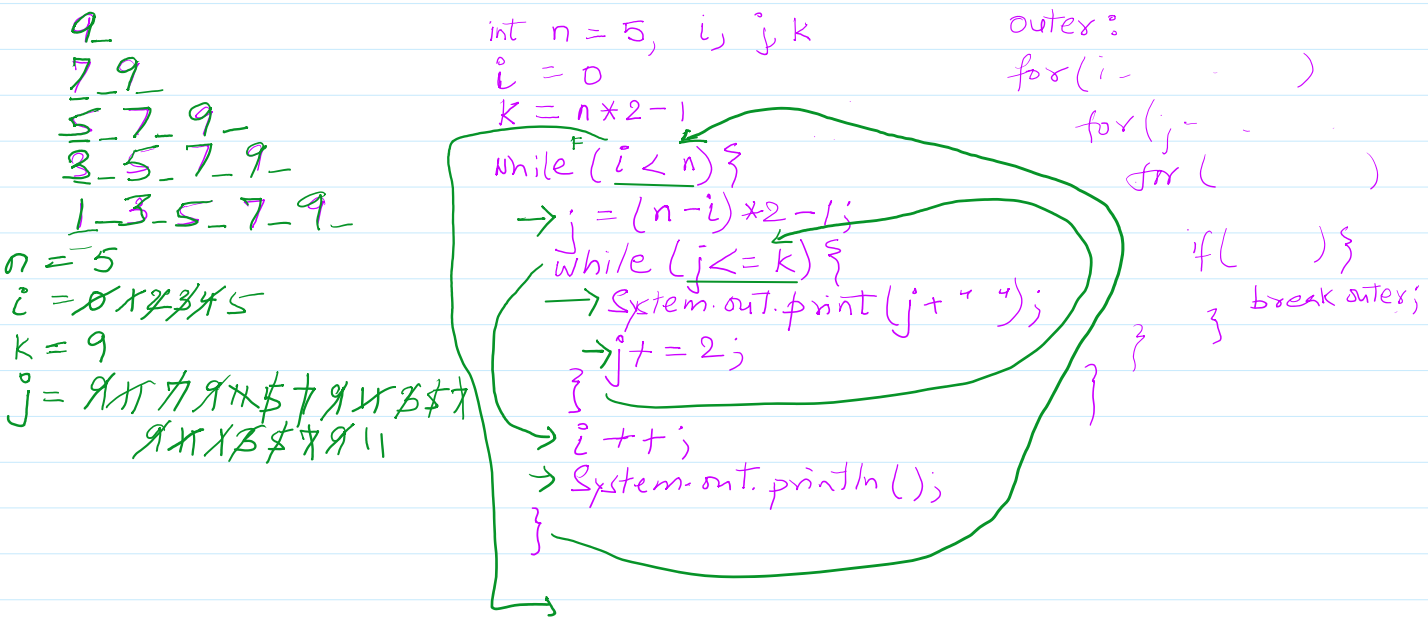

```
package NestedLoops;


/**
 * Write a description of class Q14_bPg282 here.
 * String.format("%3d", j)
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q14_bPg282
{
    public static void main(String args[]){
        System.out.print("\f");
        int row = 25;
        int i, j;
        for(i = 0; i < row; i++){
            j = 1;
            while( j <= row - i ){
                System.out.print(j+" ");
                j++;
            }
            System.out.print("\n");
        }
    }
}
```






j >= (n-i)
5-4
System.out.println();


9
7 9
5 7 9
3 5 7 9
1 3 5 7 9
n = 5
i = 0 1 2 3 4 5
k = 9
int n = 5, i, j, k
i = 0
k = n*2-1
while (i < n) {
j = (n-i)*2-1;
while (j<=k) {
System.out.print(j+" ");
j+=2;
}
i++;
System.out.println();
}
outer:
for(i
for(j
for(
if( ) }
break outer;